

Solution Requirements

1. Introduction

The Smart Lender system is a Machine Learning-based web application designed to automate the loan eligibility prediction process. The system analyzes applicant information and predicts whether a loan application should be approved or rejected. It reduces manual effort, improves prediction accuracy, and supports faster decision-making for banks and financial institutions.

2. Functional Requirements

The system should:

- Provide a user-friendly web interface.
 - Accept applicant information through an online form.
 - Validate user inputs before prediction.
 - Preprocess the input data.
 - Load the trained machine learning model.
 - Predict loan eligibility instantly.
 - Display the prediction result.
 - Handle invalid inputs gracefully.
-

3. Non-Functional Requirements

The system should provide:

- High prediction accuracy.
 - Fast response time.
 - Reliable performance.
 - Easy maintenance.
 - Scalability for future enhancements.
 - Secure handling of applicant data.
 - Browser compatibility.
 - User-friendly interface.
-

4. User Requirements

Applicant

- Enter personal and financial information.

- Receive instant loan eligibility prediction.

Bank Officer

- Verify prediction results.
- Support loan approval decisions.

Administrator

- Maintain the application.
 - Update machine learning models.
 - Monitor system performance.
-

5. Data Requirements

The system requires the following applicant details:

- Gender
 - Married Status
 - Dependents
 - Education
 - Self Employed Status
 - Applicant Income
 - Co-applicant Income
 - Loan Amount
 - Loan Amount Term
 - Credit History
 - Property Area
-

6. Machine Learning Requirements

The project includes:

- Data preprocessing
- Missing value handling
- Label encoding
- Feature scaling
- SMOTE dataset balancing
- Model training

- Model evaluation
- Cross-validation
- Model serialization using Pickle

Algorithms used:

- Decision Tree
- Random Forest
- K-Nearest Neighbors (KNN)
- Gradient Boosting (XGBoost implementation)

The Random Forest model is selected as the final deployed model based on its performance.

7. Software Requirements

- Python 3.x
 - Flask
 - Scikit-learn
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - Imbalanced-learn (SMOTE)
 - Pickle
 - HTML
 - CSS
 - Visual Studio Code
-

8. Expected Outcome

The Smart Lender system enables financial institutions to automate loan eligibility prediction, reduce manual processing time, improve prediction accuracy, and support consistent lending decisions through machine learning.

9. Conclusion

The Smart Lender solution satisfies the functional, technical, and business requirements necessary for an intelligent loan eligibility prediction system. It provides an efficient and reliable decision-support platform for modern banking applications.